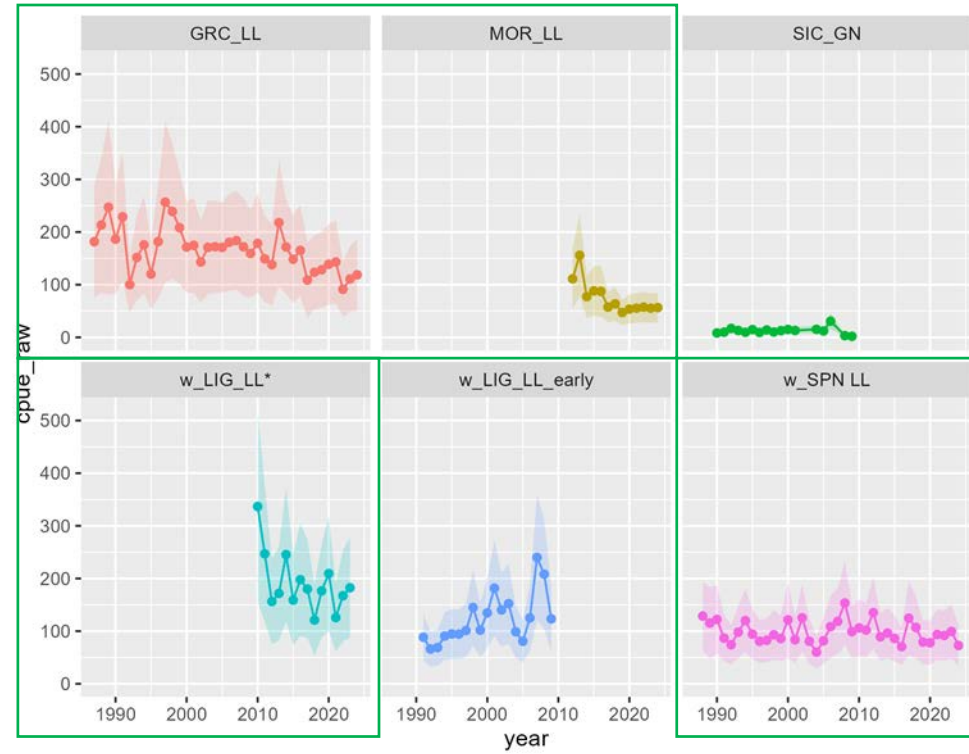
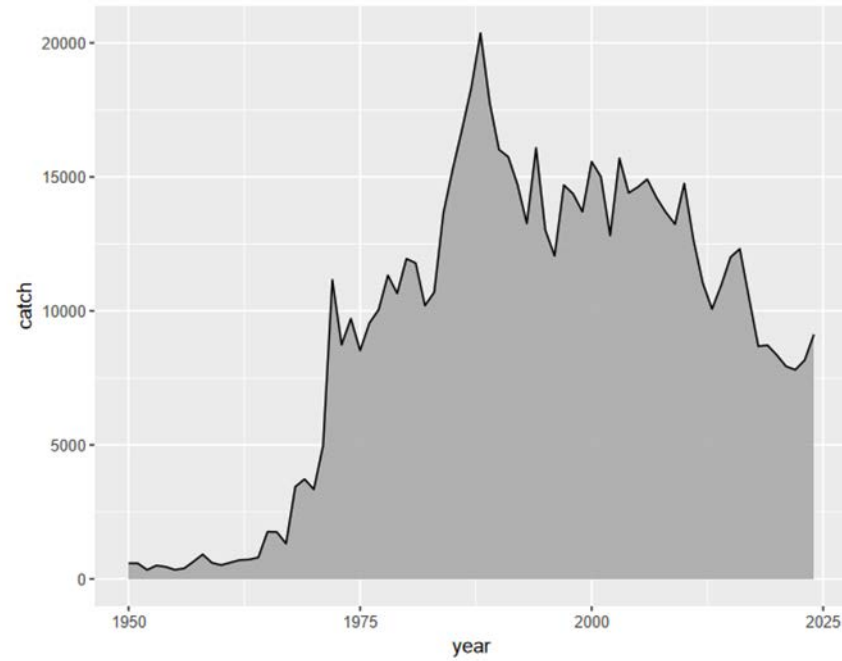


SCRS/2026/171

ASSESSMENT OF MEDITERRANEAN SWORDFISH (*Xiphias gladius*) STATUS USING BAYESIAN STATE-SPACE SURPLUS PRODUCTION MODEL JABBA

Gonzalo Liniers, Lucía Rueda, Ai Kimoto, Mauricio Ortiz, Kyle Gillespie and George Tserpes

Data used



Global Model configuration

- R package 'JABBA' v.2.3.5 (Winker et al. 2026)
- unfished biomass (K): vaguely informative lognormal prior centered at eight times the maximum historical catch, with a coefficient of variation (CV) of 1 (100%)
- initial depletion (B_{1950}/K), a lognormal prior with mean 0.95 and CV equal to 0.1 (10%)
- random catch variability on log scale set to a value of 0.2.
- Catchability parameters were formulated as uniform priors and observation variances were estimated by index assuming inverse-gamma priors.
- minimum fixed observation error equal to 0.001.
- process error variance estimated using an non-informative inverse-gamma prior with both scaling parameters set to 0.001.

Model runs

I. Continuity runs:

a. 2020 Schaefer Reference run

b. 2020 ASEM run

	Schaefer	Pella-ASEM
mu.r	0.47	0.18
cv.r	0.46	0.20
BmsyK	0.50	0.38

II. Main runs:

I. Schaefer: For the Schaefer model, the r prior distribution from the 2020 assessment was used (Winker et al. 2020b, Anom, 2020).

II. Pella-ASEM: new r prior distribution and the associated parameter (m) were calculated using an Age-Structured Equilibrium Model (ASEM) approach (Winker et al., 2020a).

III. Pella- m : conducted using a Pella-Tollimson production function with the shape parameter m estimated by the JABBA model and with a default CV on the shape parameter equal to 0.3.

III. Sensitivity runs (we'll see them later)

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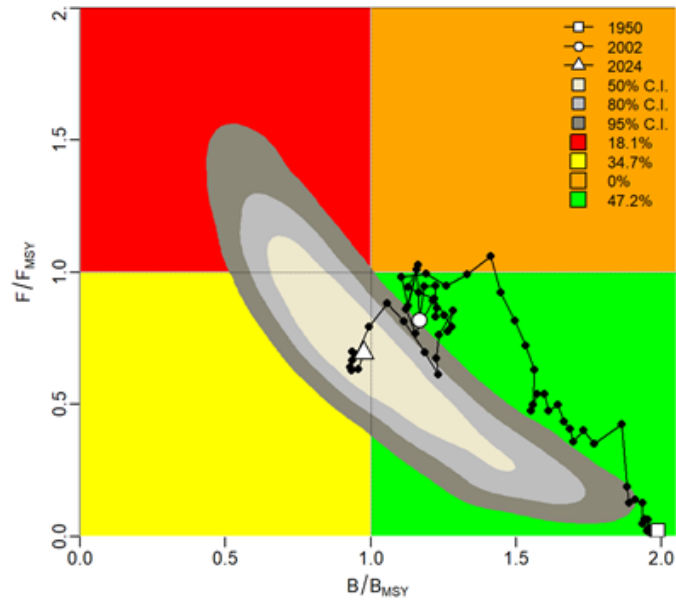
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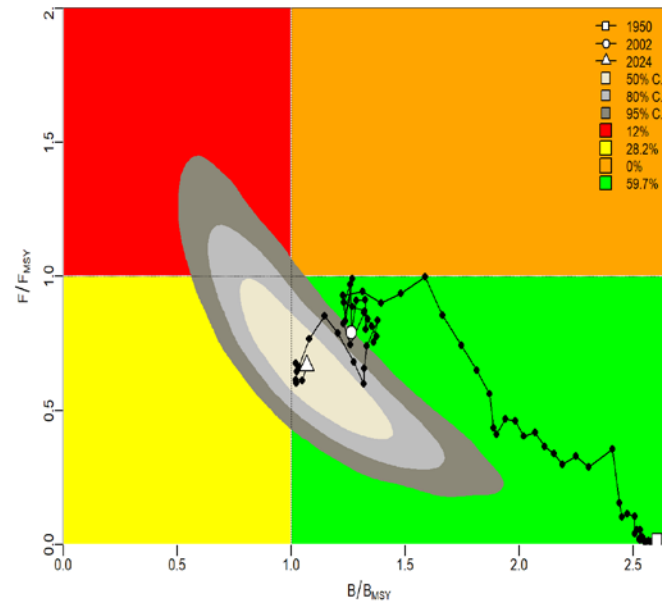
III. Sensitivity runs (we'll see them later)

Results → Continuity runs

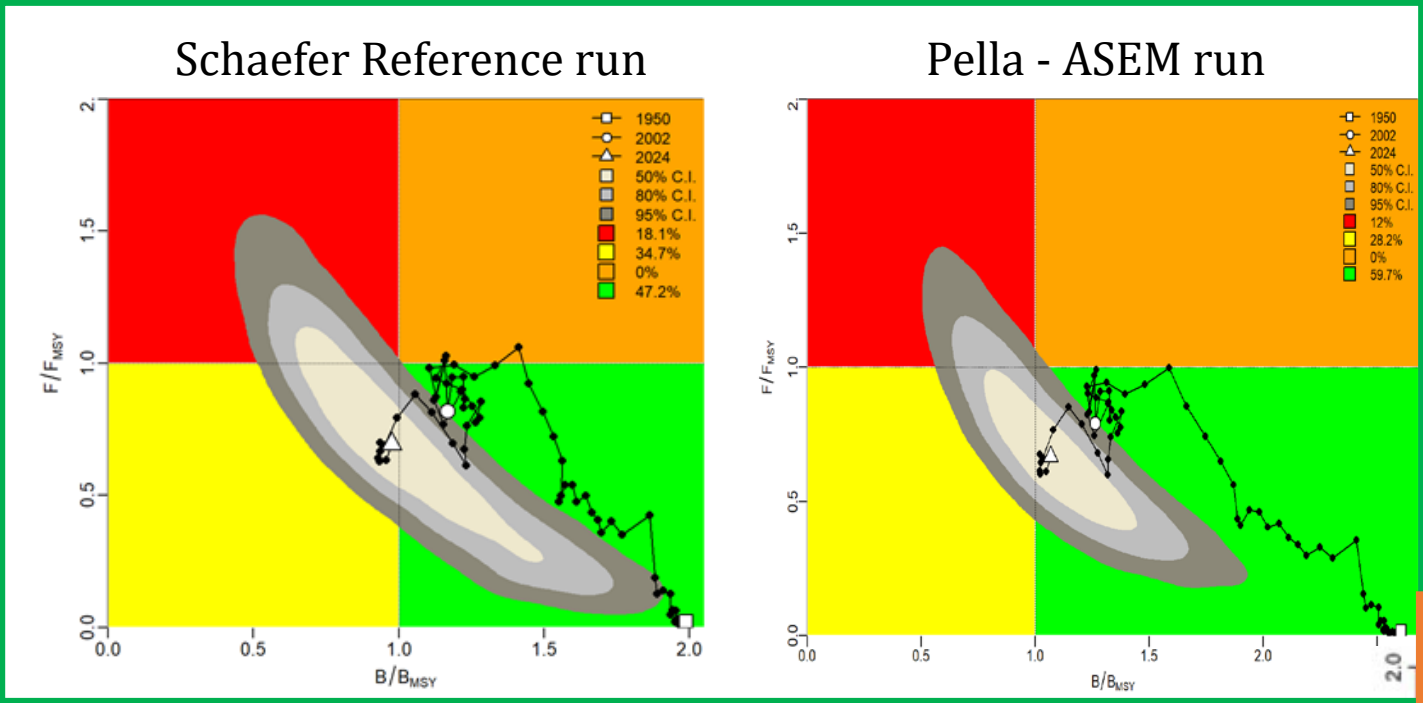
Schaefer Reference run



Pella - ASEM run

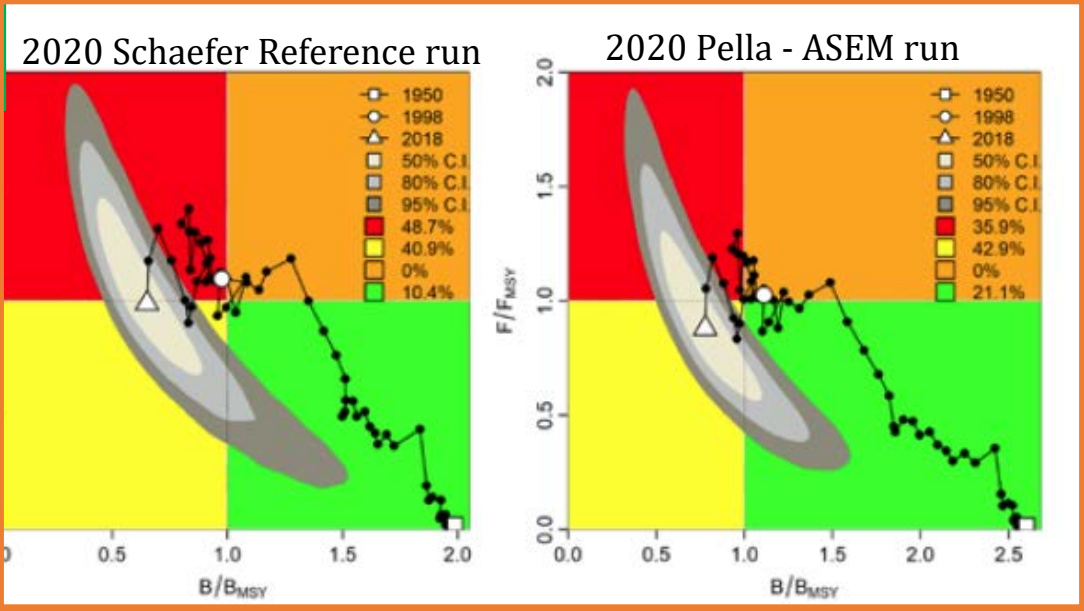


Results -> Continuity runs



2026

2020

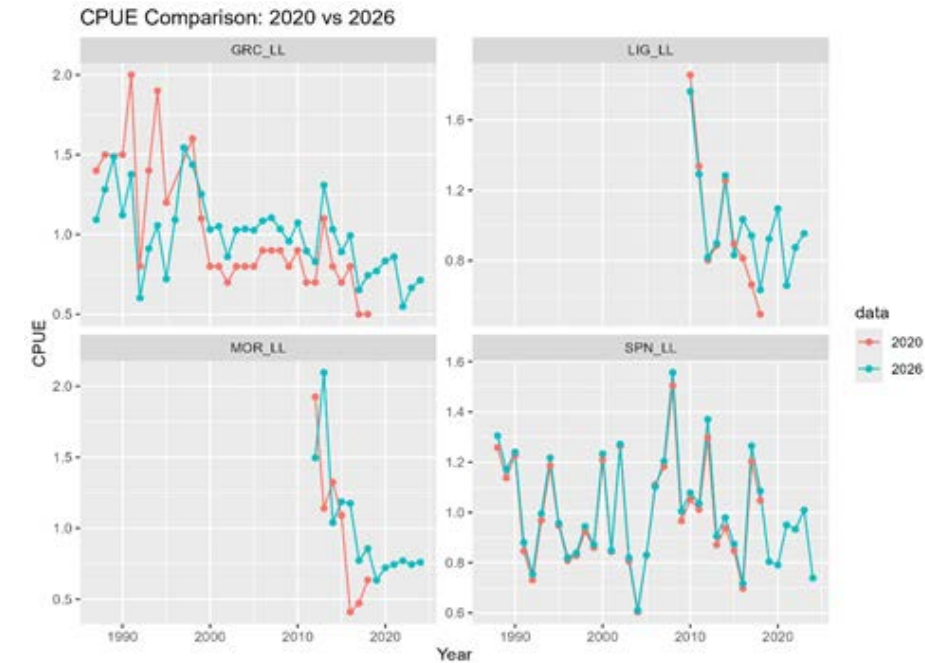
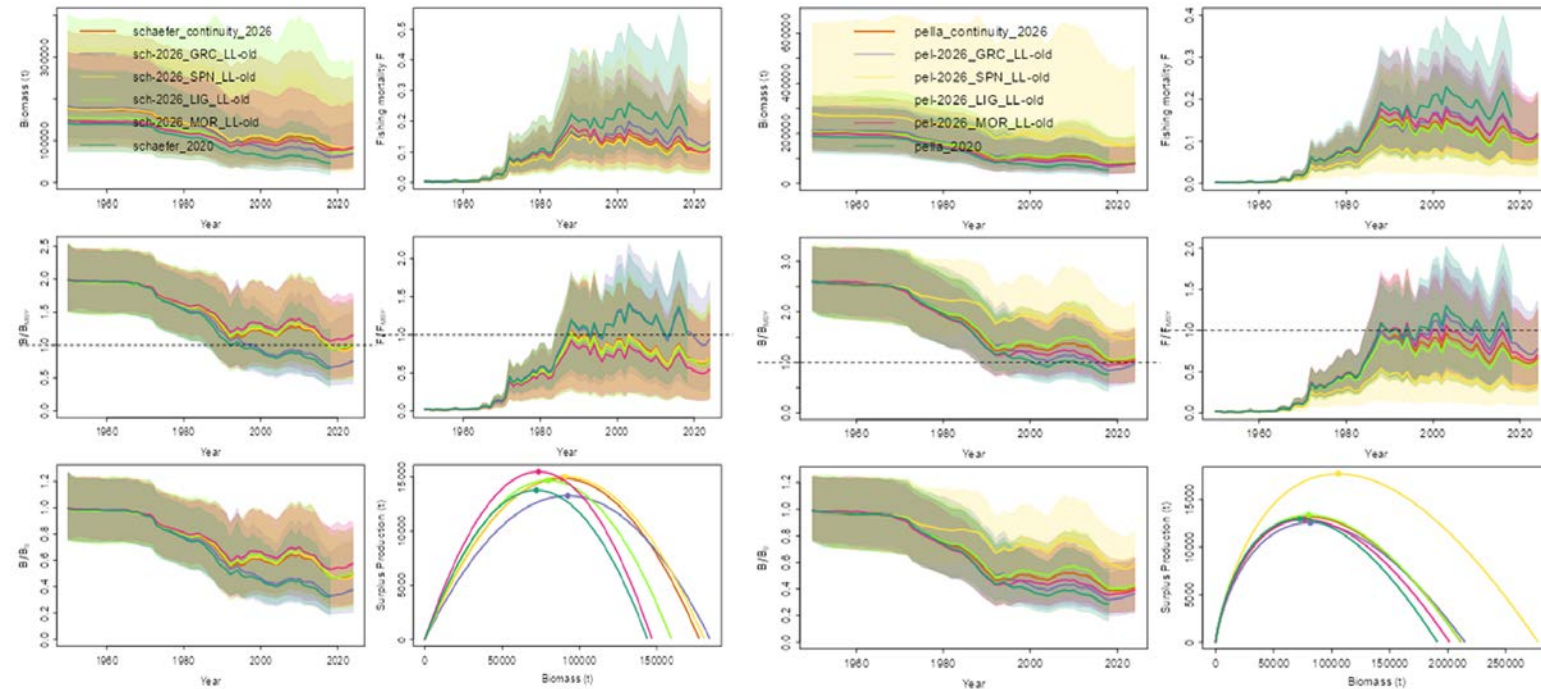


Results -> Continuity runs -> Sensitivity runs

- (i) Jackknife index analysis
- (ii) use of a fixed CV equal to 0.2 for all years and all indices (similar to the 2020 assessment)
- (iii) fitting models up to 2018 (last year of data of the 2020 assessment) but using the relative abundance indices provided for the present assessment
- (iv) re-fitting the continuity runs but using the old abundance indices at a time

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III. Sensitivity runs (we'll see them later)

Results -> Main runs -> 2026 ASEM

<i>Parameter</i>	<i>Mean</i>	<i>CV</i>	<i>Distribution</i>	<i>Description</i>	<i>Source</i>
M	0.206	0.25	lognormal	Natural mortality (1/year)	McAllister (2014)
L _{inf}	238.58	0.1	lognormal	Von Bertalanffy asymptotic length	Mean: ICCAT Manual CV: Working Group
K	0.185	0.1	normal	Von Bertalanffy growth parameter	Mean: ICCAT Manual CV: Working Group
t ₀	-1.404	0.2	normal	Von Bertalanffy age at zero length	Mean: ICCAT Manual CV: Working Group
a	9.61E-06	-	exponential	Weight at length parameter (GG-LJFL)	Tserpes et al. (2017)
b	3.059	-	non lognormal	Weight at length parameter (GG-LJFL)	Tserpes et al. (2017)
L ₅₀	136.5	0.2	lognormal	Length at 50% maturity	Mean: SCRS/2026/055 Manual CV: McAllister (2014)
d	0.2	0.2	lognormal	Parameter of the logistic maturity ogive	Working Group
h	0.83	0.14	beta	Steepness $h=0.2 + 0.8$ Beta (5.86. 1.59)	McAllister (2014)

- (i) slope parameter of L_{50} (dL_m) equal to 1/0.2;
- (ii) maximum age equal to 15 years;
- (iii) sequence of range of F values for iterative MSY estimation from 0.01 to 0.9 each 0.002;
- (iv) length at 50% retention (SL_{50}) equal to 100 cm based on the analysis from the cumulative frequency distribution from the Spanish measured lengths.

Results -> Main runs -> 2026 ASEM

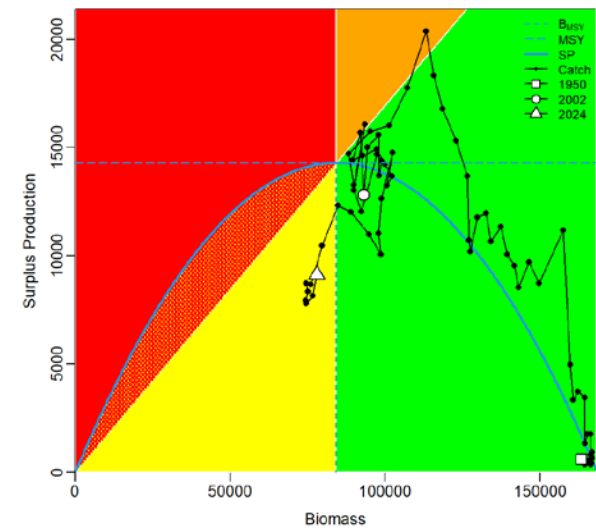
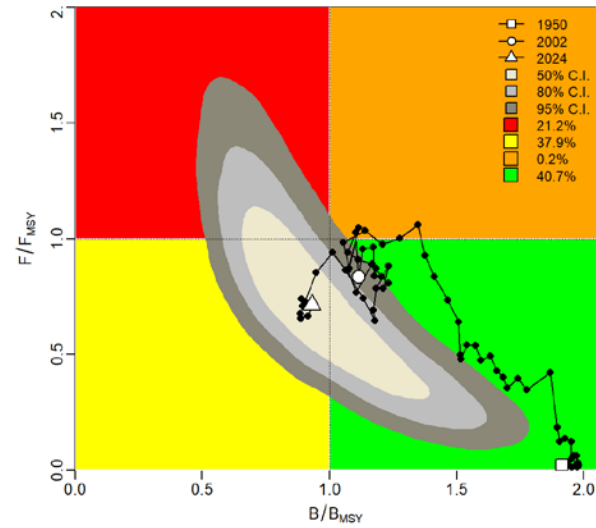
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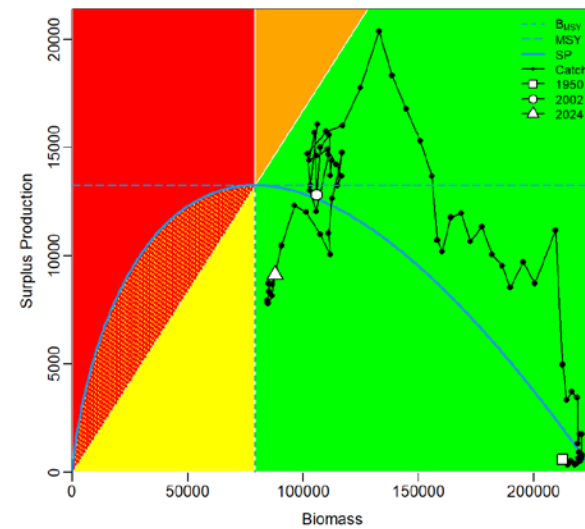
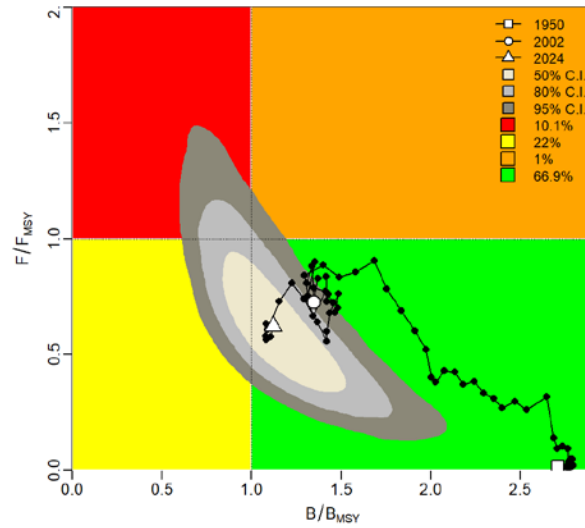
mu r	0.154
CV r	0.096
Bmsy/B0	0.35

Results -> Main runs

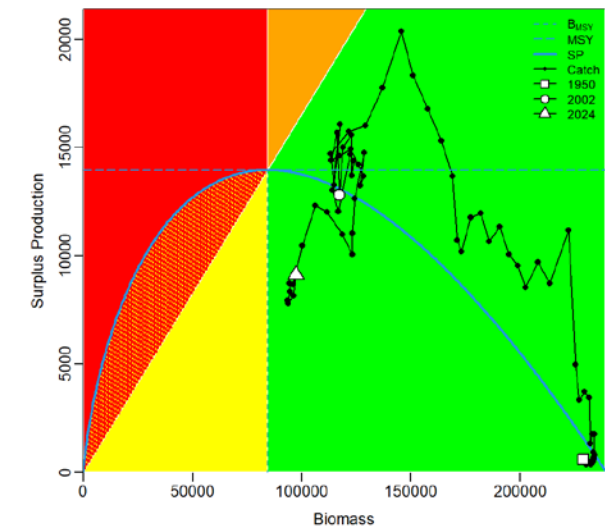
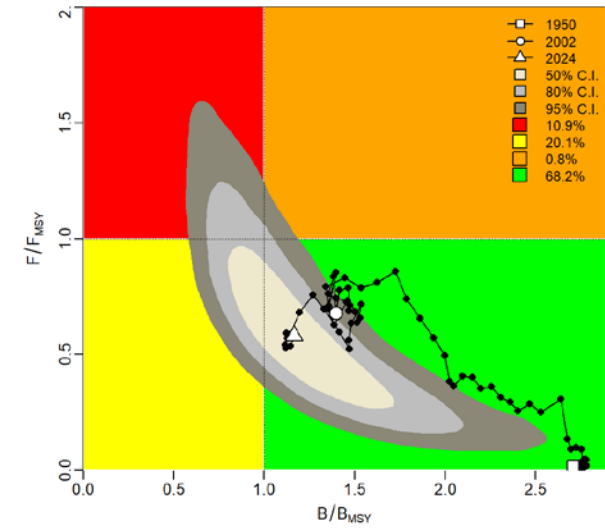
Schaefer



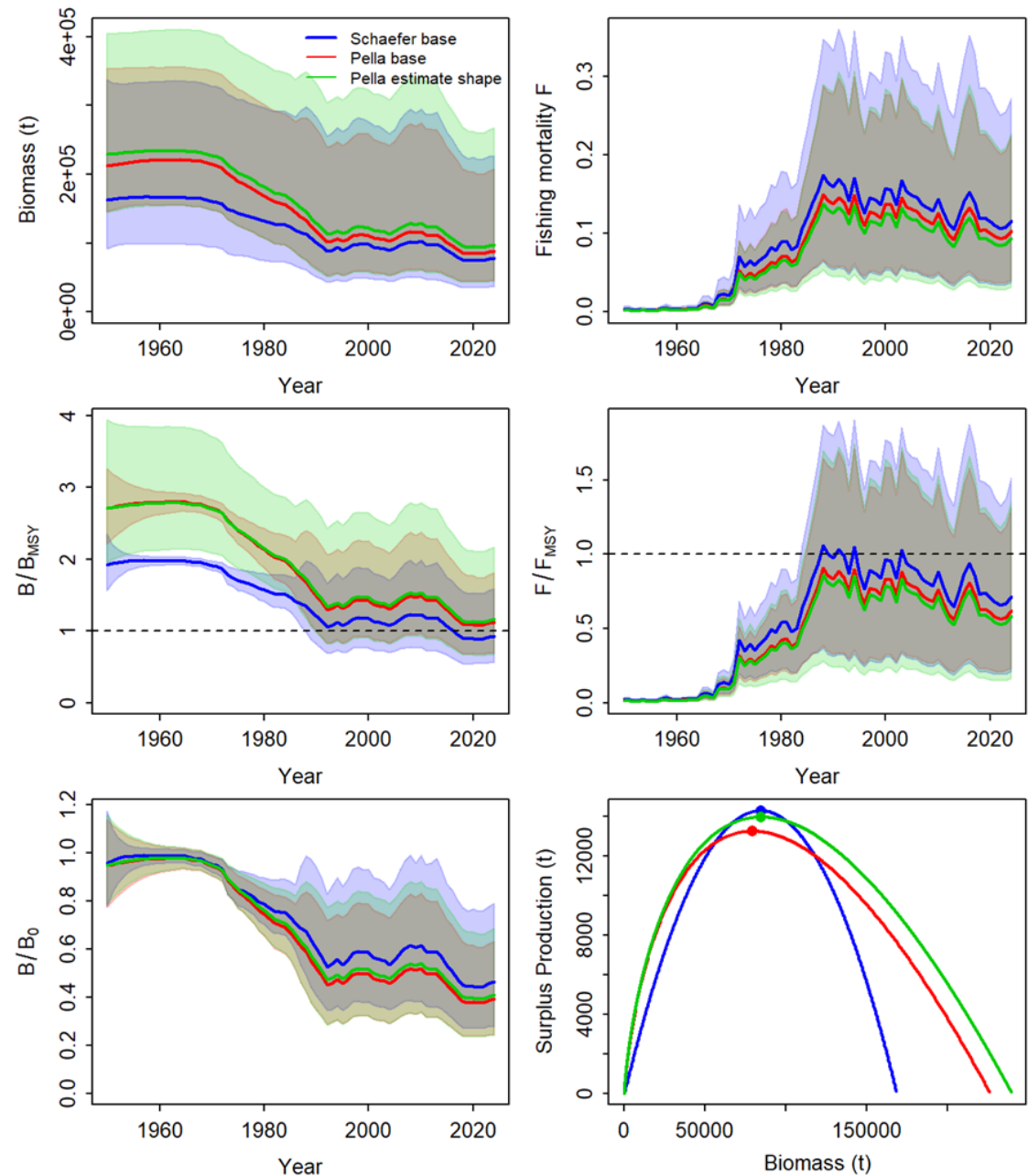
Pella-ASEM



Pella-m



Results -> Main runs



<i>Model</i>	<i>quant</i>	<i>K</i>	<i>B_{MSY}</i>	<i>F_{MSY}</i>	<i>MSY</i>
Schaefer	median	168,509	84,254	0.1695	13,723
Pella ASEM	median	226,319	79,229	0.1672	13,081
Pella-Est_m	median	239,562	84,495	0.1655	13,414

Results → Main runs

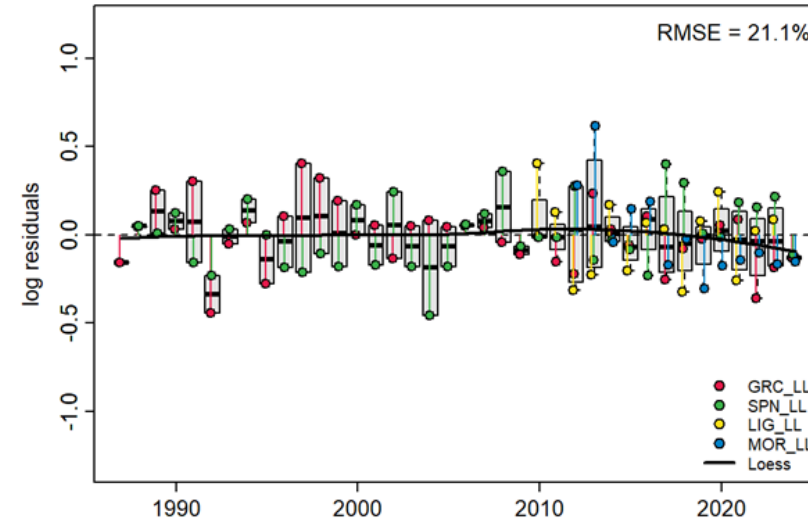
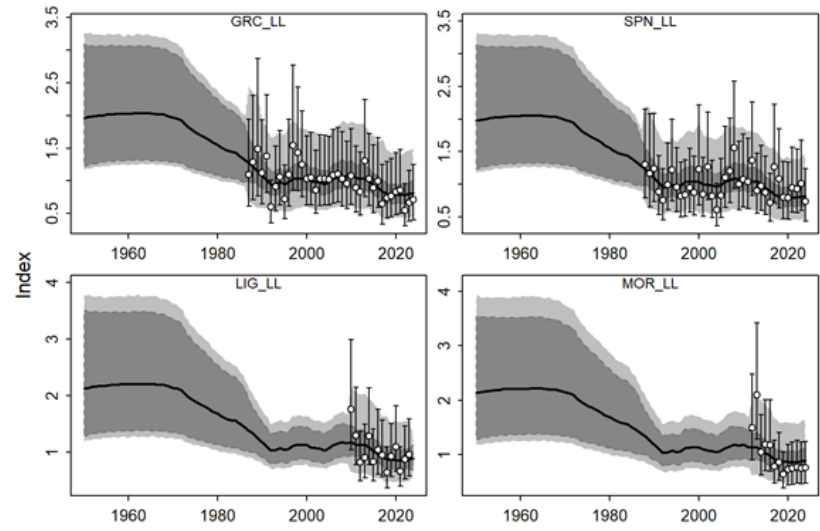
Schaefer			
	Median	LCI	UCI
K	1.685092e+05	9.963645e+04	3.380386e+05
r	3.390181e-01	1.792374e-01	6.794201e-01
q.1	1.060000e-05	3.700000e-06	2.120000e-05
q.2	1.070000e-05	3.700000e-06	2.150000e-05
q.3	1.150000e-05	3.800000e-06	2.410000e-05
q.4	1.150000e-05	3.900000e-06	2.430000e-05
psi	9.578258e-01	7.806876e-01	1.175400e+00
sigma2	7.786900e-03	2.059500e-03	2.345230e-02
tau2.1	2.917600e-03	4.221000e-04	2.000470e-02
tau2.2	3.060400e-03	4.659000e-04	2.228390e-02
tau2.3	6.551500e-03	7.049000e-04	6.632640e-02
tau2.4	8.858500e-03	5.540000e-04	1.013433e-01
m	2.000000e+00	2.000000e+00	2.000000e+00

Pella ASEM			
	Median	LCI	UCI
K	2.263195e+05	1.589714e+05	3.621131e+05
r	1.516330e-01	1.069767e-01	2.177541e-01
q.1	9.300000e-06	4.000000e-06	1.710000e-05
q.2	9.400000e-06	4.000000e-06	1.740000e-05
q.3	1.010000e-05	4.200000e-06	1.930000e-05
q.4	1.010000e-05	4.300000e-06	1.940000e-05
psi	9.484221e-01	7.749153e-01	1.140076e+00
sigma2	7.063600e-03	2.350400e-03	2.026990e-02
tau2.1	3.121100e-03	4.749000e-04	2.008580e-02
tau2.2	3.179100e-03	4.915000e-04	2.222130e-02
tau2.3	6.194900e-03	5.820000e-04	6.731020e-02
tau2.4	8.766400e-03	5.612000e-04	1.005923e-01
m	9.070000e-01	9.070000e-01	9.070000e-01

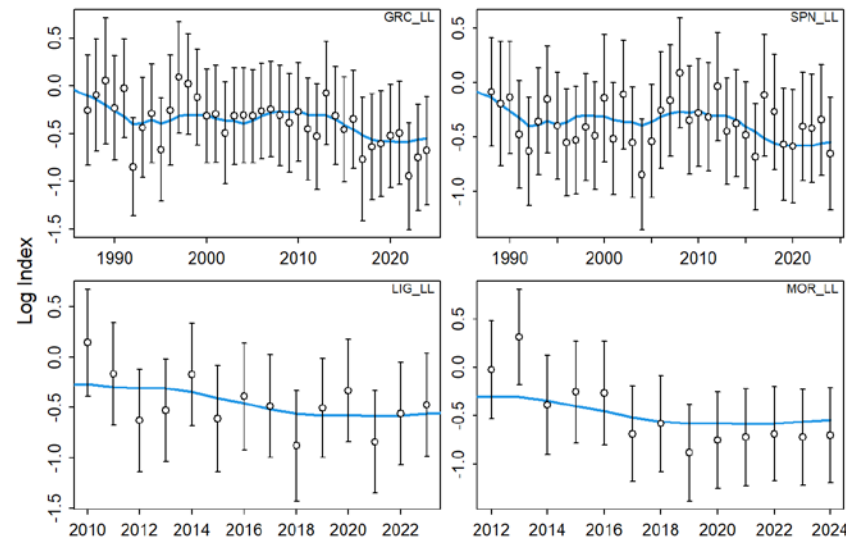
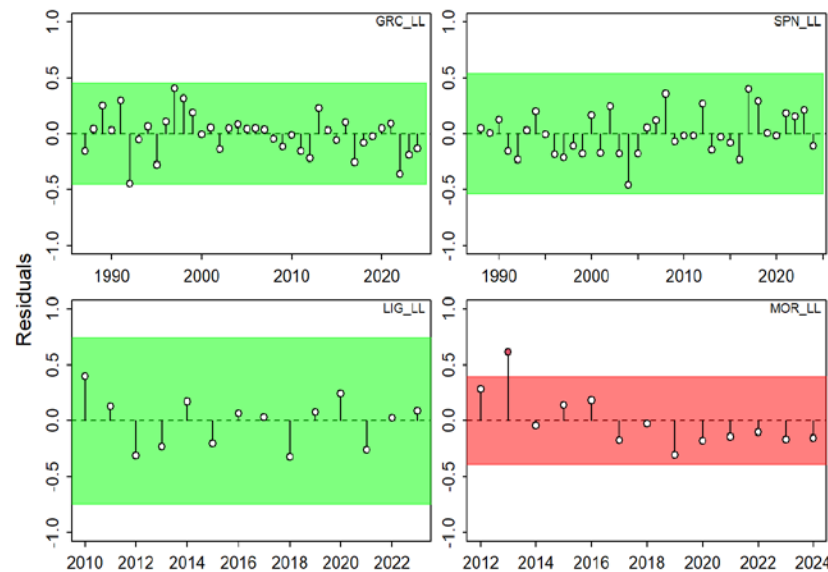
Pella-Est_m			
	Median	LCI	UCI
K	2.395624e+05	1.586851e+05	4.187666e+05
r	1.517561e-01	1.054065e-01	2.178550e-01
q.1	8.400000e-06	3.200000e-06	1.760000e-05
q.2	8.500000e-06	3.200000e-06	1.780000e-05
q.3	9.100000e-06	3.300000e-06	1.980000e-05
q.4	9.100000e-06	3.300000e-06	1.990000e-05
psi	9.508470e-01	7.900412e-01	1.158195e+00
sigma2	6.774900e-03	2.175600e-03	1.962540e-02
tau2.1	2.936400e-03	4.385000e-04	2.026450e-02
tau2.2	3.211100e-03	4.486000e-04	2.223880e-02
tau2.3	5.867300e-03	6.463000e-04	6.399070e-02
tau2.4	9.217700e-03	6.326000e-04	1.034056e-01
m	9.172791e-01	5.190161e-01	1.603323e+00

Results -> Main runs -> Diagnostics

CPUE fits and residuals



* Only Pella-ASEM shown here, other models very similar results

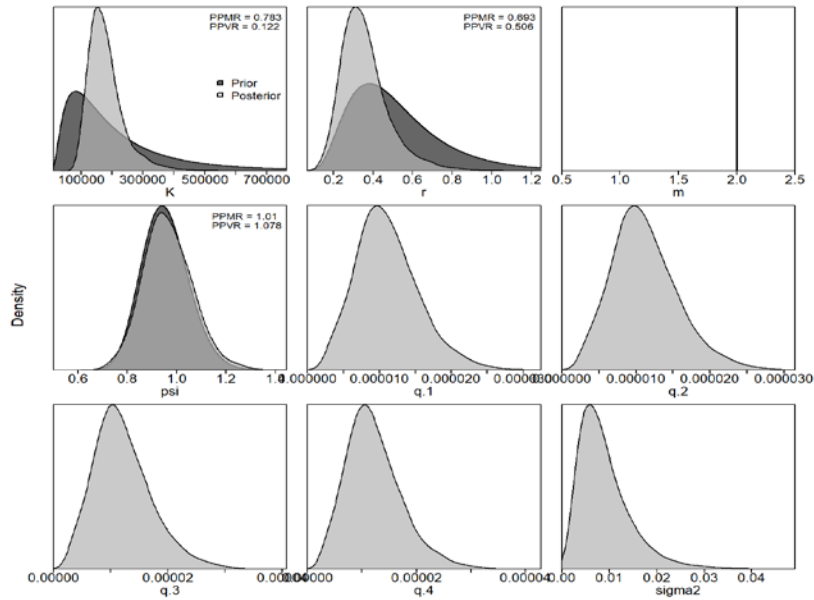


13-17 July 2020)

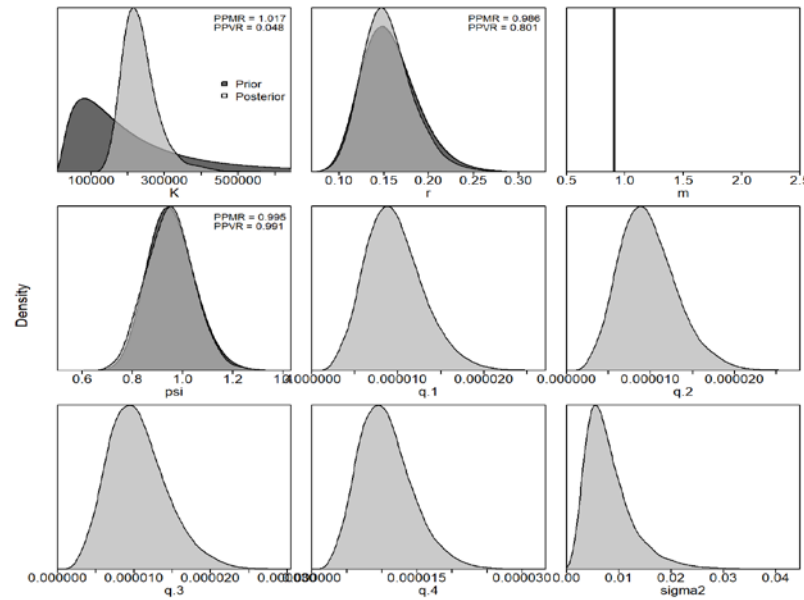
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Prior and posterior distributions for the parameters

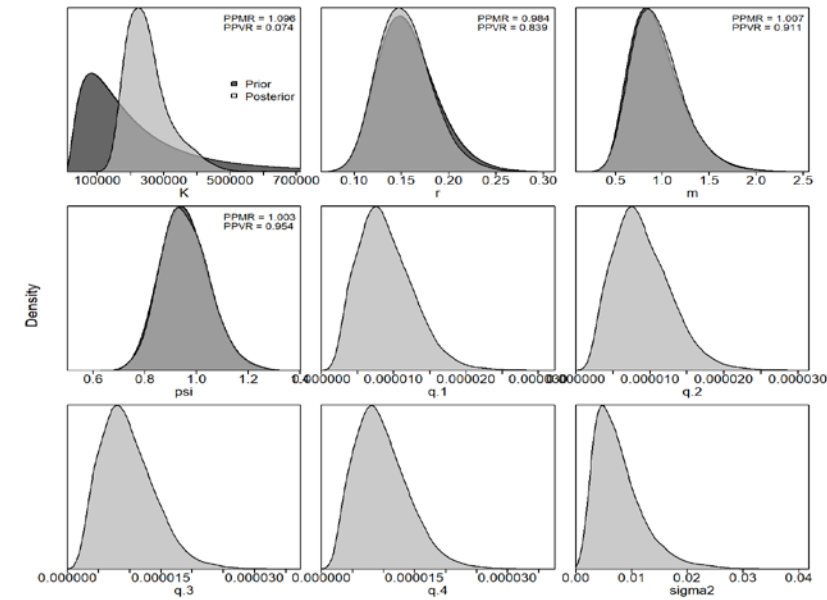
Schaefer



Pella-ASEM



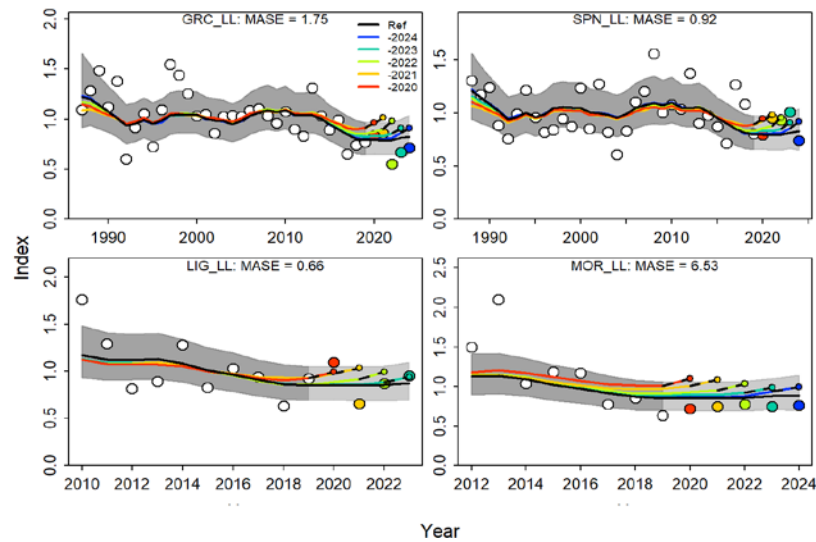
Pella-m



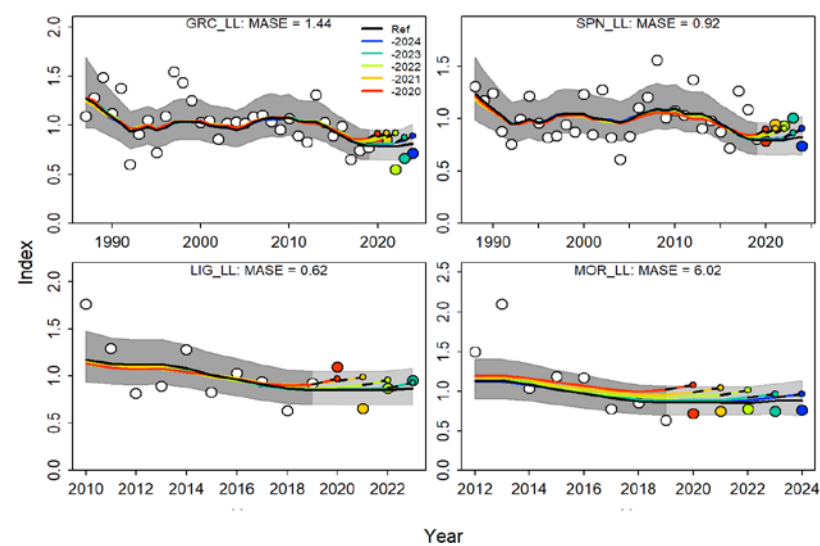
Results -> Main runs -> Diagnostics

Hindcast analysis

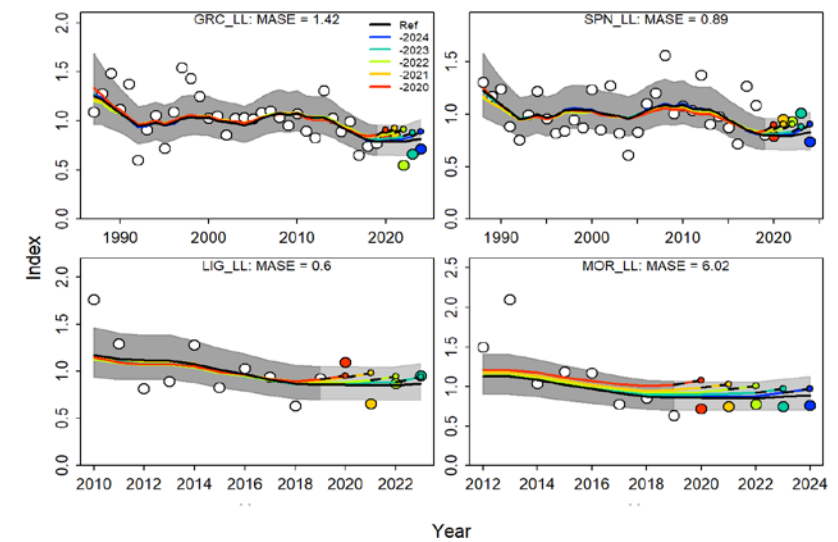
Schaefer



Pella-ASEM



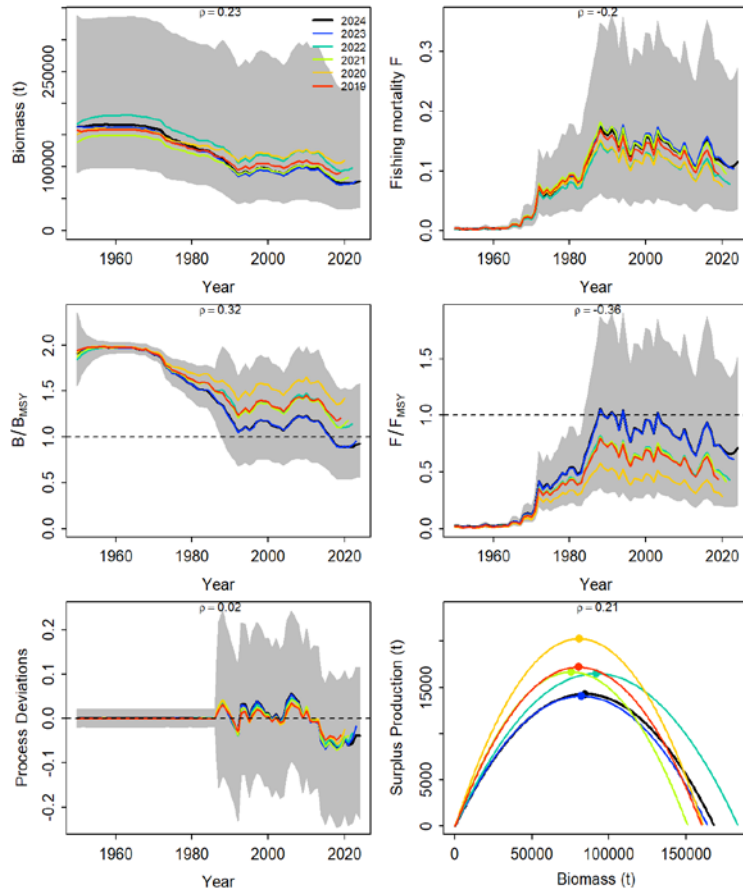
Pella-m



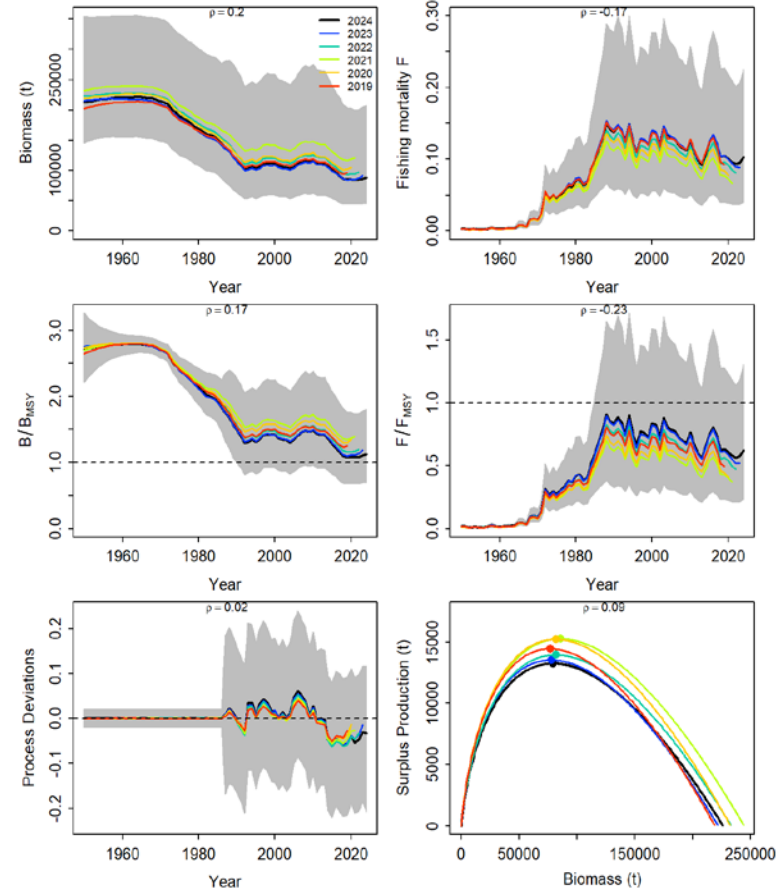
Results -> Main runs -> Diagnostics

Retrospective analysis

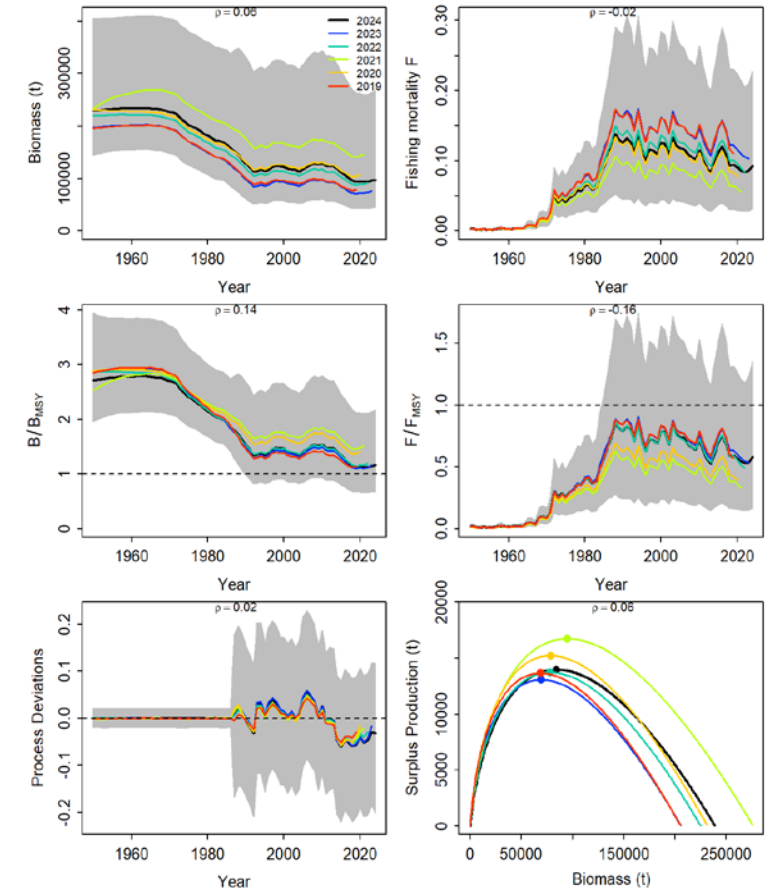
Schaefer



Pella-ASEM



Pella-m



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III. Sensitivity runs

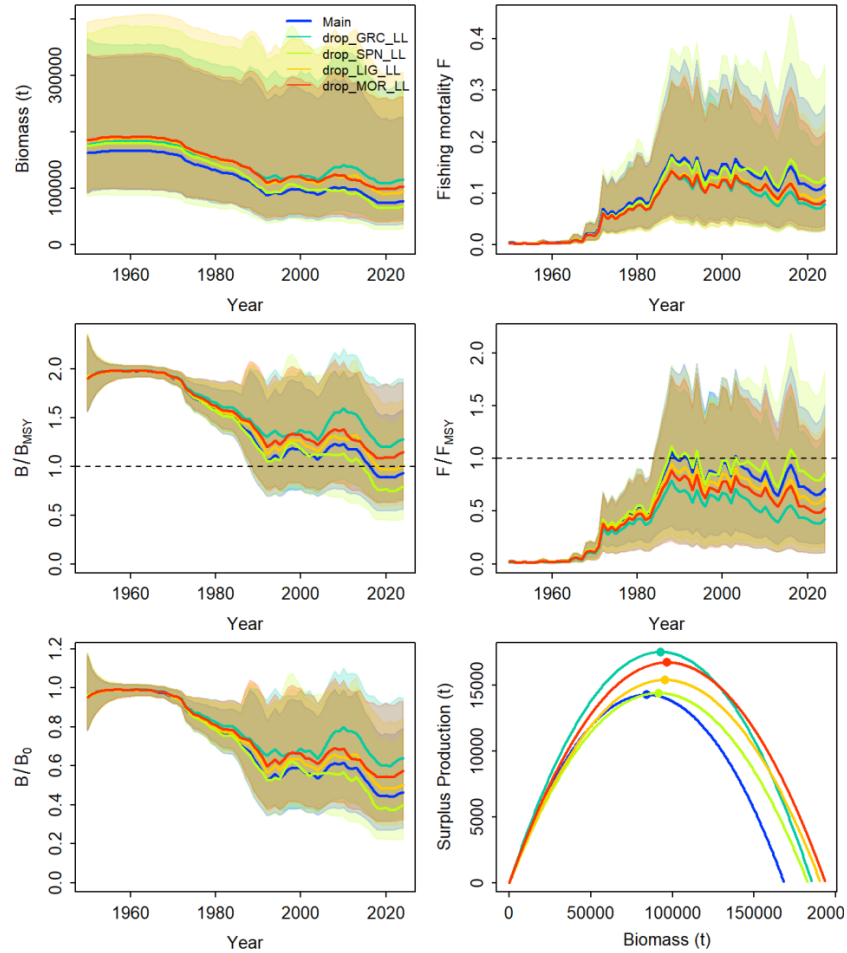
Sensitivity analysis

- i. Jackknife index analysis consisting on re-fitting the Main runs excluding one of the four abundance indices used in the Main run at a time.
- ii. Under-reporting catches using alternative catches scenarios by scaling the catches from 1950 to 1986 by factors of 1.2, 1.5, 2 and 3 and re-fitting the Main model with the inflated historical catches.
- iii. set the initial biomass depletion (B_{1950}/K) prior mean to 0.85 .
- iv. run with no random variation on catches.
- v. Use historical indices of abundance (SIC_GN and early_LIG_LL)

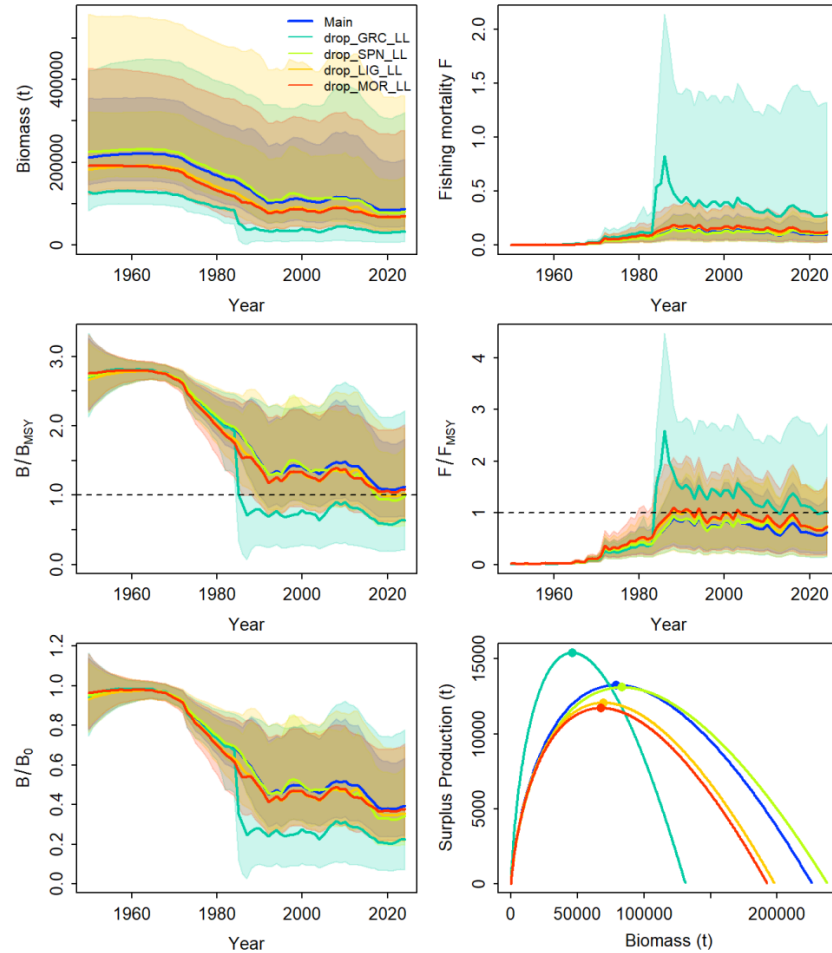
Results -> Sensitivity analyses

(i) Jackknife index analysis

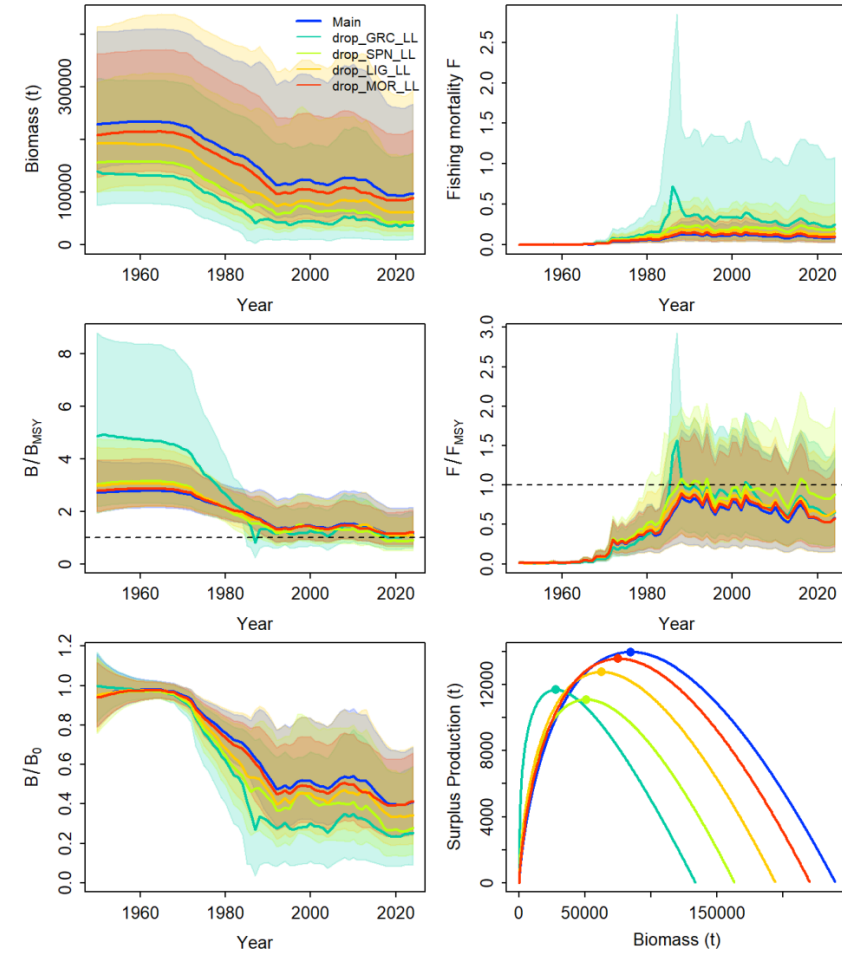
Schaefer



Pella-ASEM

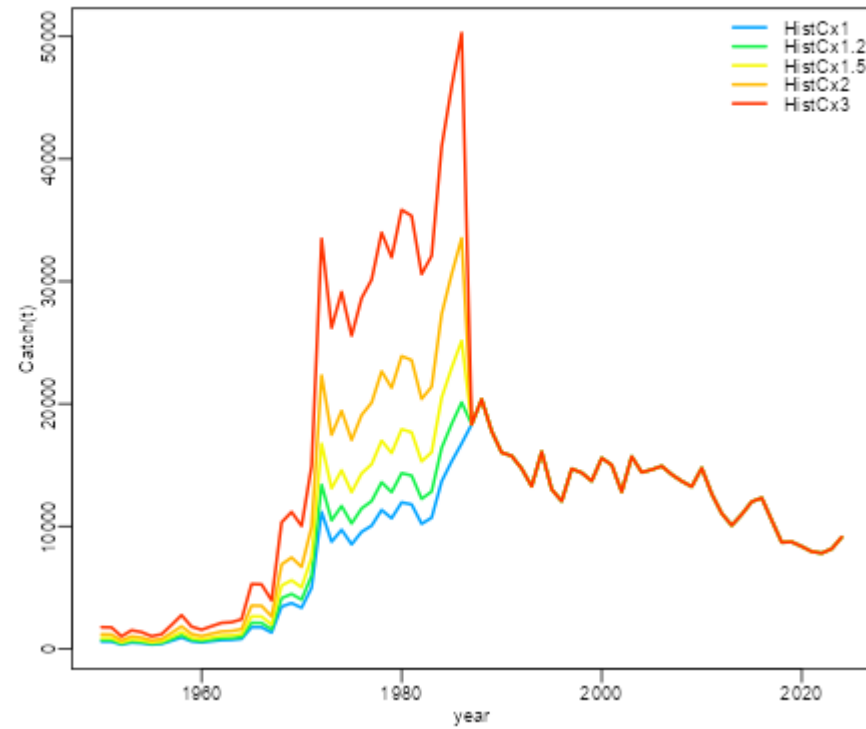


Pella-m



Results -> Sensitivity analyses

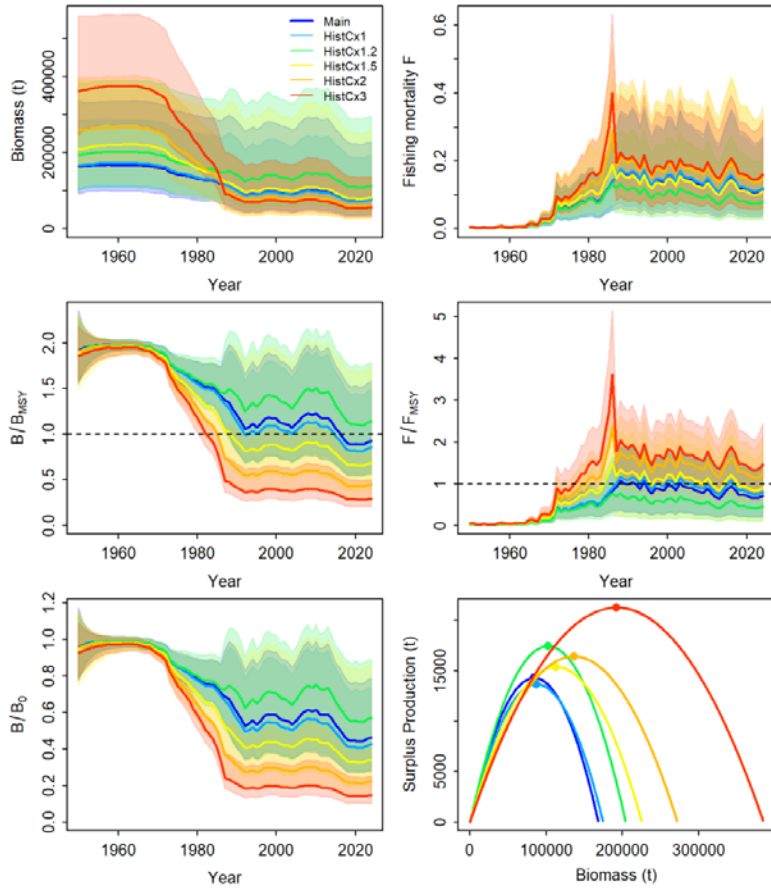
(ii) Under-reporting catches



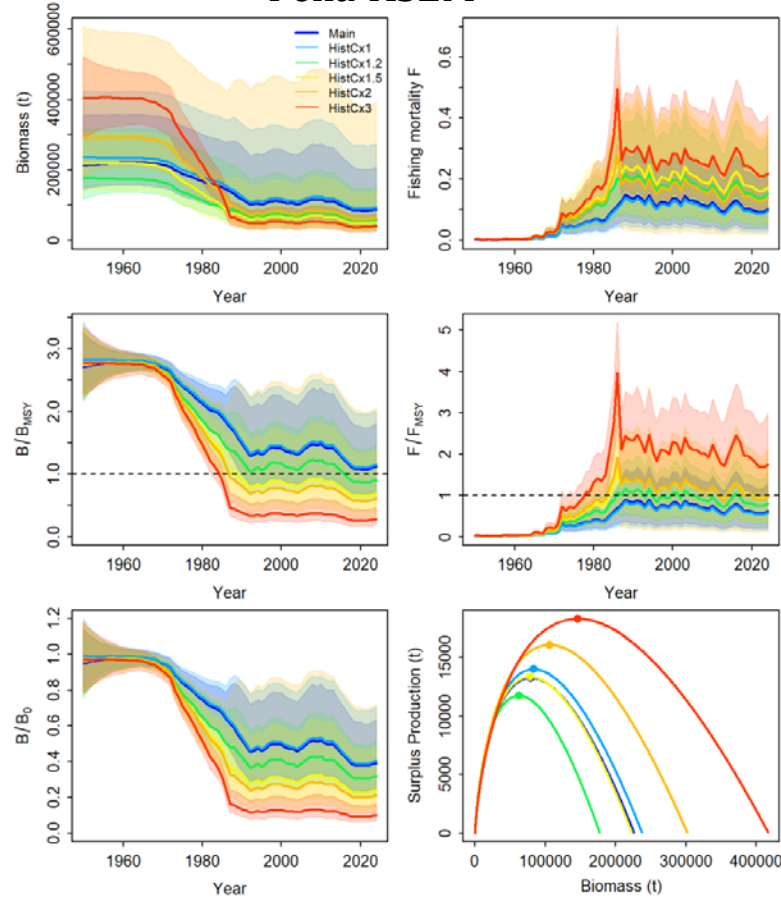
Results -> Sensitivity analyses

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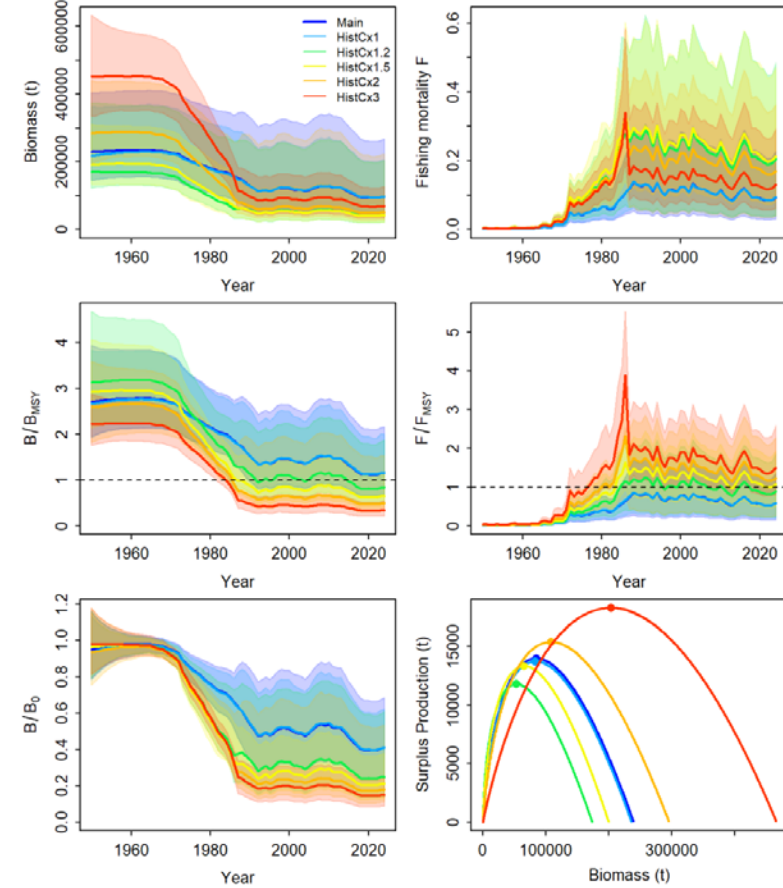
Schaefer



Pella-ASEM



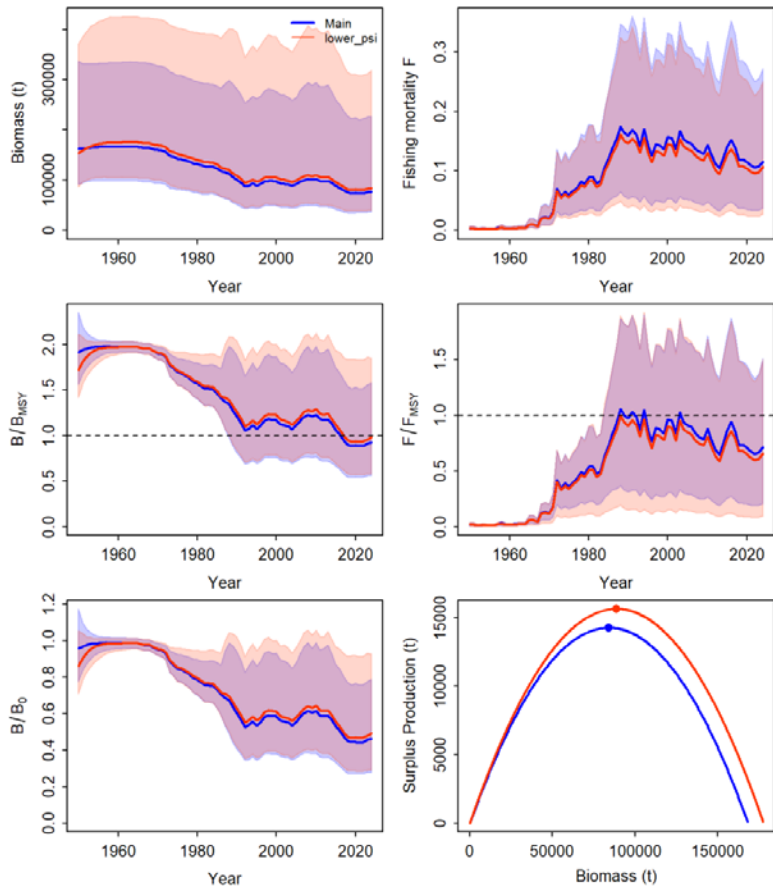
Pella-m



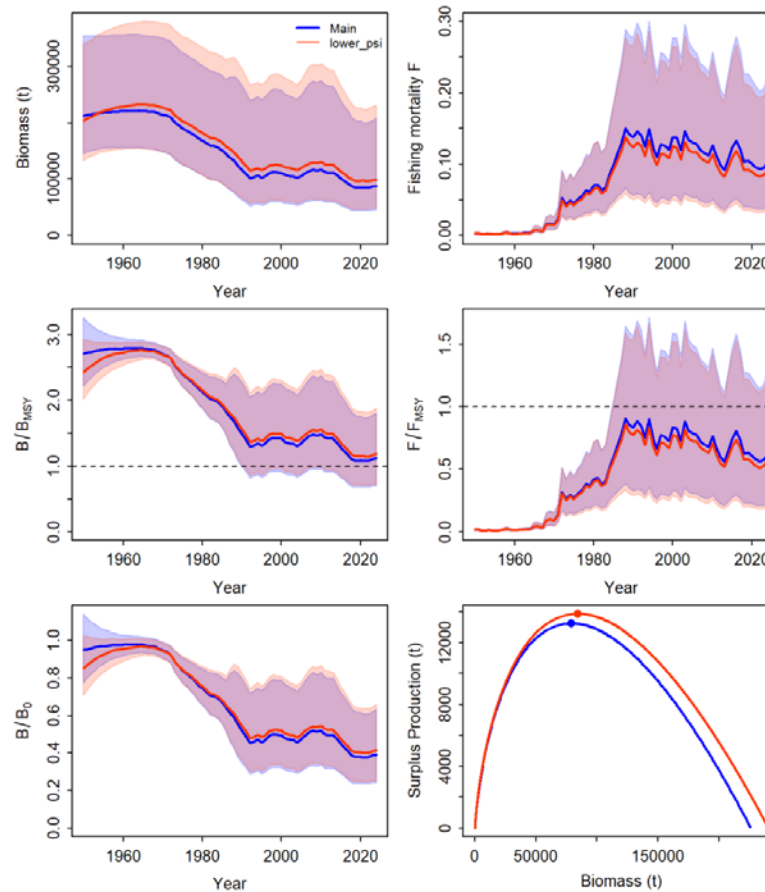
Results -> Sensitivity analyses

(iii) Initial depletion 0,85

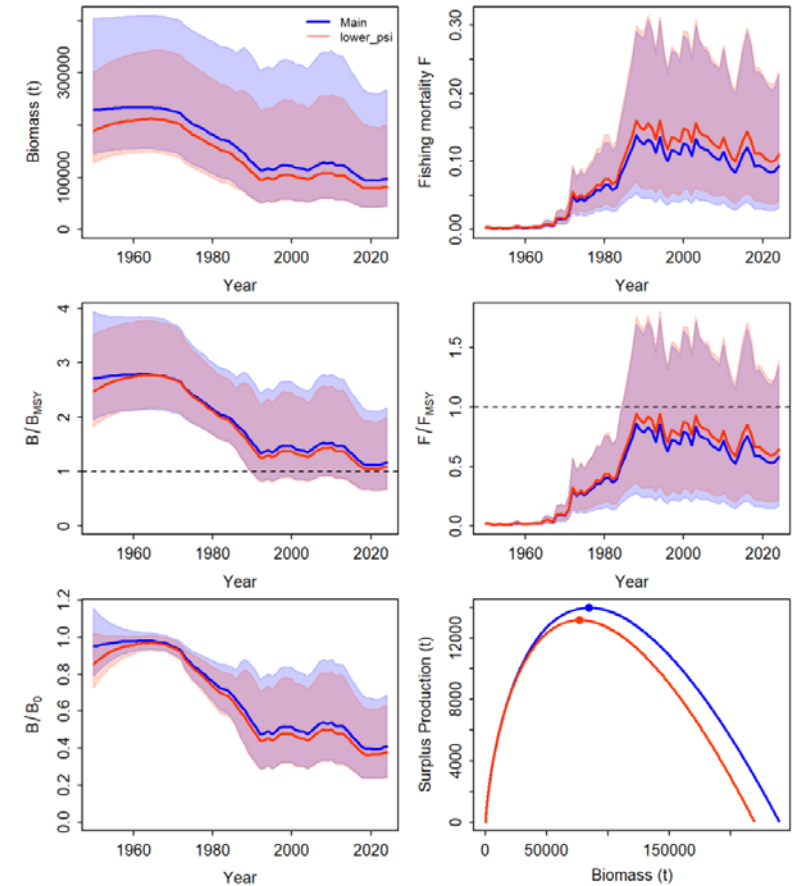
Schaefer



Pella-ASEM



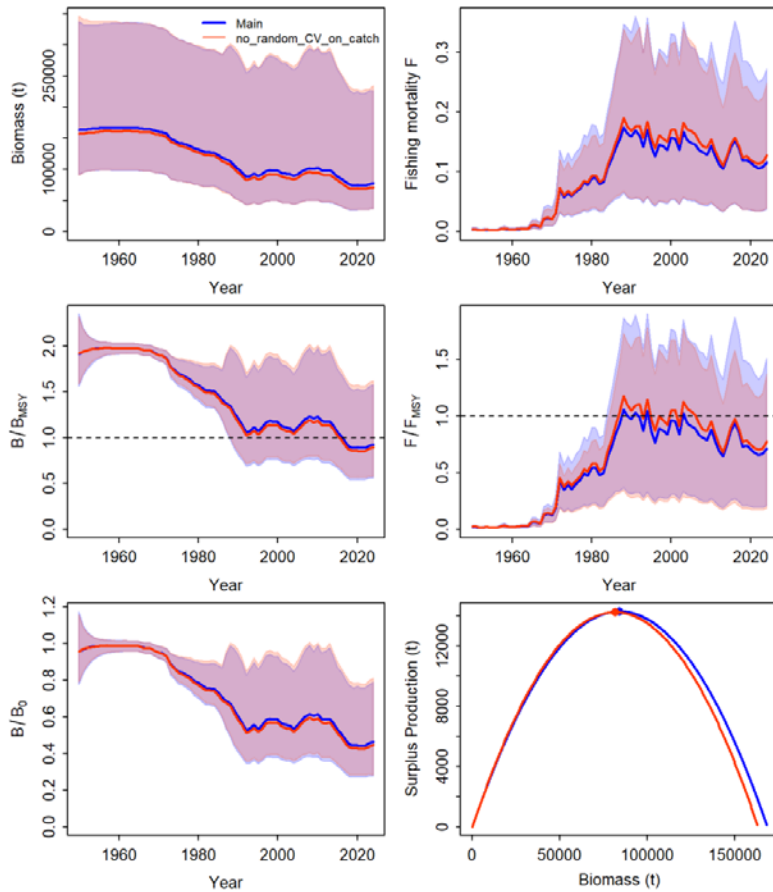
Pella-m



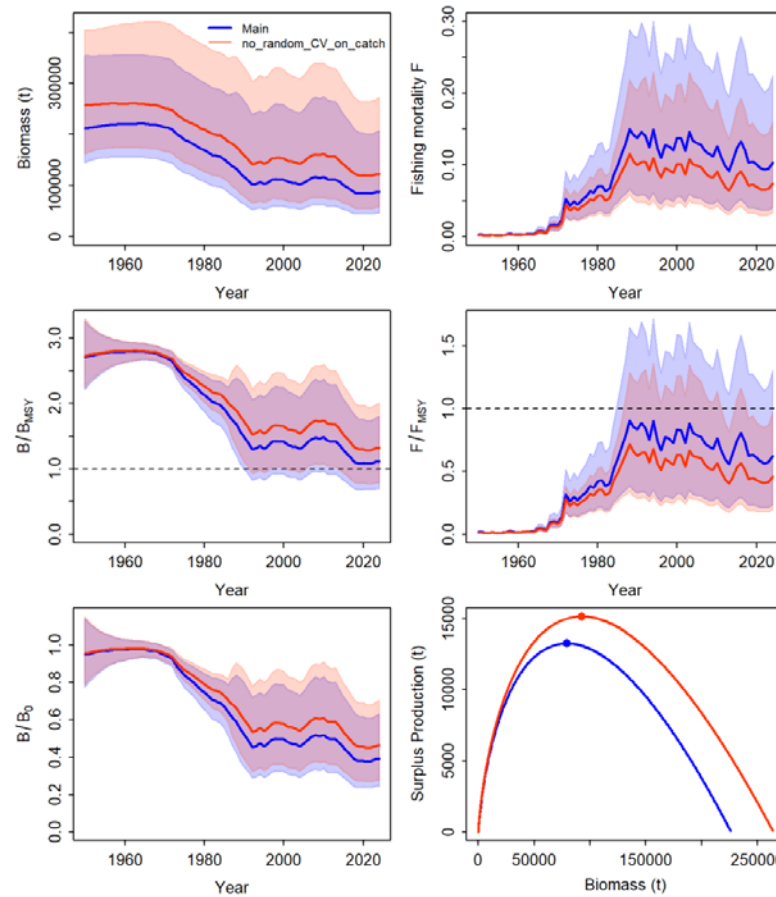
Results -> Sensitivity analyses

(iv) No random variation on catches

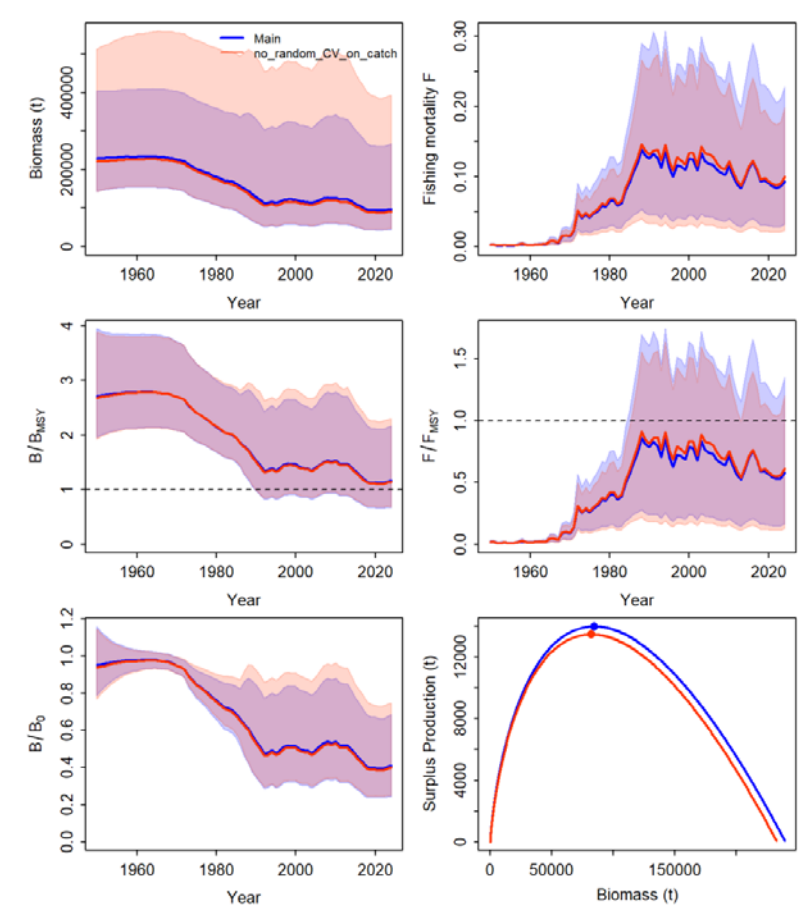
Schaefer



Pella-ASEM

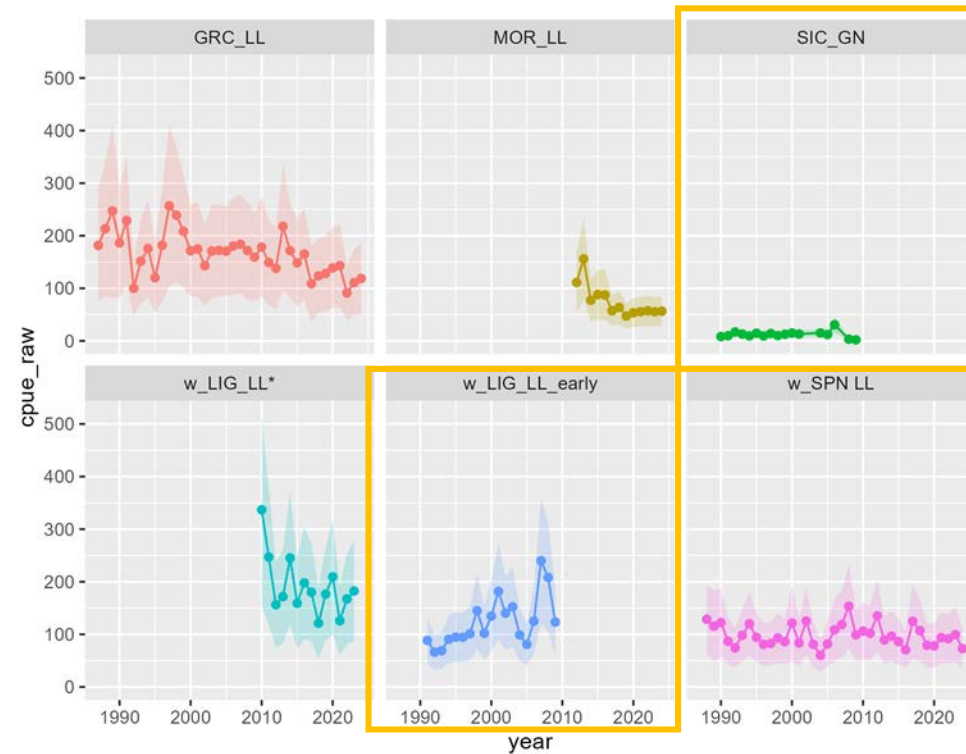


Pella-m



Results -> Sensitivity analyses

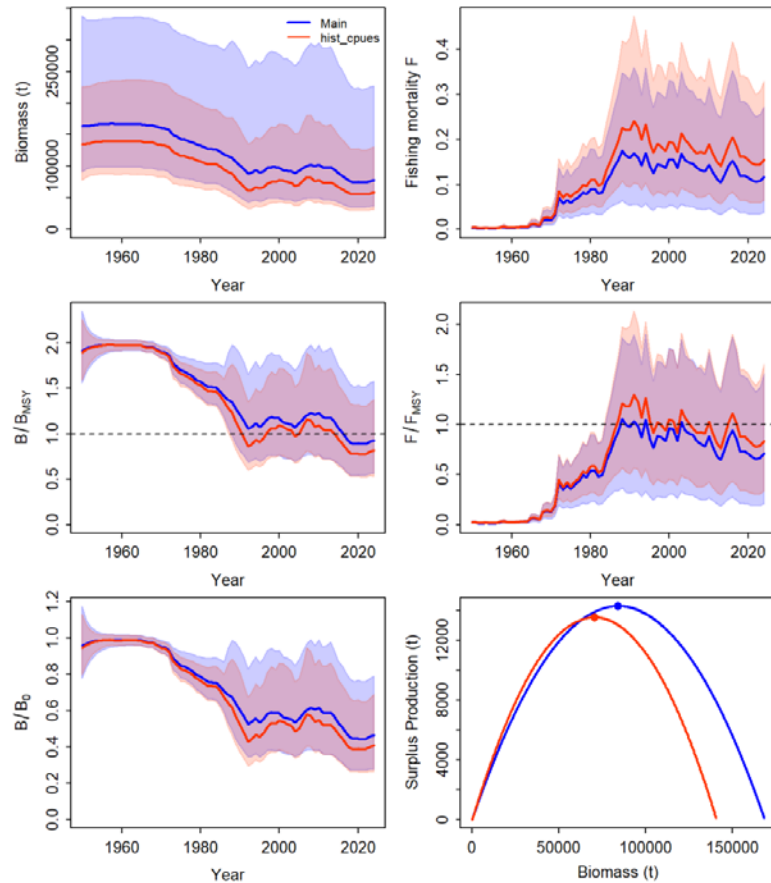
(v) historical indices of abundance



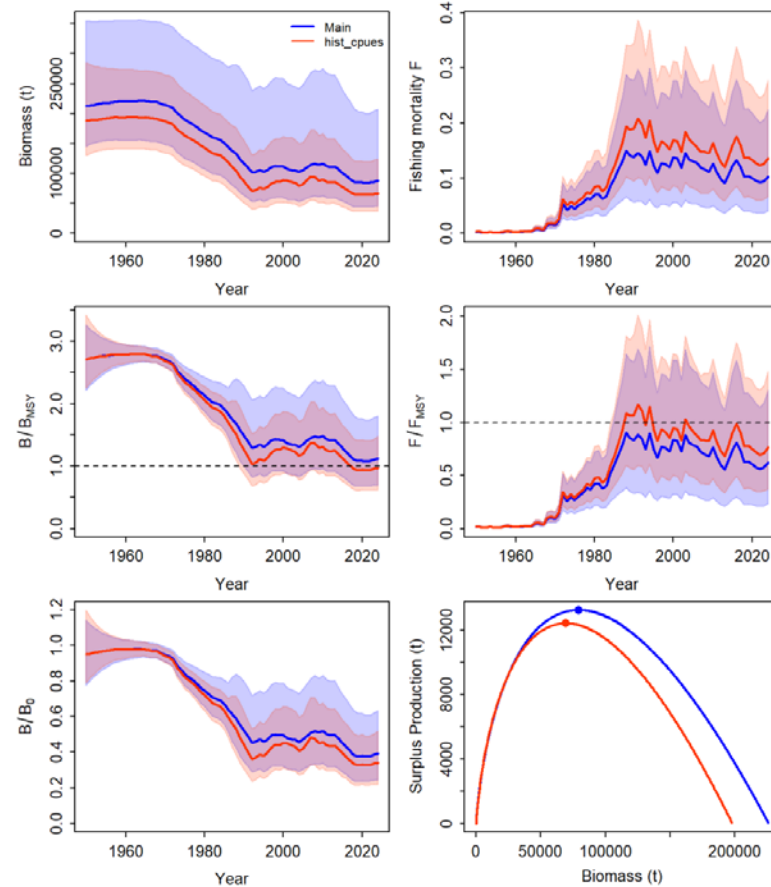
Results → Sensitivity analyses

(v) historical indices of abundance

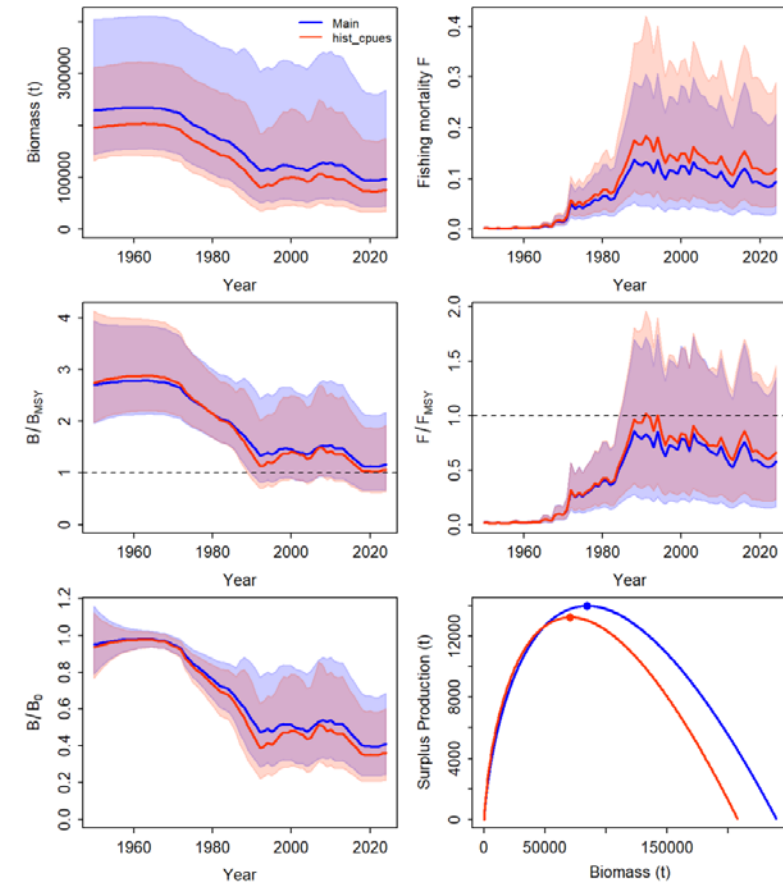
Schaefer



Pella-ASEM



Pella-m



GRACIAS

This work has been co-funded by the European Union through the European Maritime Fisheries and Aquaculture Fund (EMFAF) within the National Program of collection, management and use of data in the fisheries sector and support for scientific advice regarding the Common Fisheries Policy.